

中国海洋大学全日制本科课程期末考试试卷

2025 年 Spring 季学期 考试科目: 统计建模与推断 学院: 海德学院

试卷类型: SAMPLE EXAM ONLY 卷 命题人: Andy RONG 审核人:

考试说明: 本课程为开卷考试, 共6页, 可携带计算器。

Question	1	2	3	4	5	Total
Marks	20	20	10	20	30	

学号: _____ 姓名: _____ 专业年级: _____ 授课教师: _____ 考场教室号: _____ 座号: _____

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Question 1 [20 Marks Total]

A toy company has developed a new electronic robot toy. We need to evaluate:

- Marketing claim about customer interest (Part 1)
- Engineering claim about product performance (Part 2)

Part 1: The marketing department claims the robot attracts an average of 3.5 young customers (ages 6-8) per 20-minute store demonstration. Store managers suspect this claim is exaggerated. Let X be the number of young customers who purchase the toy during a 20-minute demonstration. We test:

$$H_0 : \lambda = 3.5 \quad \text{vs} \quad H_a : \lambda < 3.5$$

with decision rule: Reject H_0 if $X \leq 1$.

Part 2: The engineering team claims the robot's sound module operates at exactly 2000 kHz. Eight independent tests yielded these frequencies (in kHz):

1982, 2005, 1991, 1978, 1989, 1998, 2001, 1992

Assume frequencies are normally distributed.

- (a) 4 Marks Find the probability of Type I error (α)
- (b) 4 Marks Calculate the Type II error probability (β) when the true rate is $\lambda = 2.5$
- (c) 2 Marks Determine the test's power when $\lambda = 1.5$
- (d) 4 Marks Construct a 95% confidence interval for the true mean frequency
- (e) 3 Marks Test the engineering claim at $\alpha = 0.025$ significance level
- (f) 3 Marks Find the p-value for this test

Question 2 [20 Marks Total]

A study was conducted to compare the reaction times (in milliseconds) of two groups: Group A (experienced gamers) and Group B (non-gamers). Independent random samples of 10 participants from each group were employed in the experiment. The results are shown in the table below.

Table 1: Reaction Time Data	
Group A (Gamers)	Group B (Non-Gamers)
$n_1 = 10$	$n_2 = 10$
$\bar{y}_1 = 245.3 \text{ ms}$	$\bar{y}_2 = 278.6 \text{ ms}$
$s_1^2 = 48.16$	$s_2^2 = 52.89$

(a) [20 Marks] Do the data provide sufficient evidence to suggest a difference between the true mean reaction times for the two groups? Use $\alpha = 0.05$.

Question 3 [10 Marks Total]

Suppose $Y_1, Y_2, \dots, Y_n$ are i.i.d. random variables from the normal distribution $N(\mu, \sigma^2)$. Define the sample mean and sample variance as:

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

(a) 5 Marks Show that:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

(b) 5 Marks Split the data into two independent groups:

- Group 1: $Y_1, Y_2, \dots, Y_k$
- Group 2: $Y_{k+1}, Y_{k+2}, \dots, Y_n$

Let S_1^2 and S_2^2 denote the sample variances of the two groups. Show that:

$$\frac{S_1^2}{S_2^2} \sim F_{k-1, n-k}$$

Question 4 [20 Marks Total]

The following questions are based on the R output from a linear regression model predicting city fuel efficiency (cty) using highway fuel efficiency (hwy) and number of cylinders (cyl):

```
null_model <- lm(cty ~hwy+cyl , data = mpg)
summary(null_model)
confint.lm(null_model,level=0.90)
```

Call:

```
lm(formula = cty ~ hwy + cyl, data = mpg)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.0196	-0.6022	-0.0196	0.7576	5.2402

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.07210	0.82956	7.320	4.09e-12 ***
hwy	???????	0.01947	29.924	< 2e-16 ***
cyl	-0.48757	0.07195	???????	1.01e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.146 on 231 degrees of freedom

Multiple R-squared: 0.9281, Adjusted R-squared: 0.9275

F-statistic: 1490 on 2 and 231 DF, p-value: < 2.2e-16

- 2 Marks** How many observations are in the data?
- 8 Marks** Using the R output above:
 - Calculate the estimate for hwy (β_1)
 - Calculate the t-value for $\hat{\beta}_2$ (cyl coefficient)
- 4 Marks** Find the 90% confidence interval for the intercept term
- 6 Marks** Perform a statistical test to determine if there is a significant (linear) relationship between cty and the two predictors (hwy, cyl) at the 10% significance level. State:
 - Null and alternative hypotheses
 - Test statistic value
 - P-value
 - Conclusion

Question 5 [30 Marks Total]

A study investigated the growth characteristics of blue mussels in New South Wales, with samples from two locations. The data includes age (weeks) and weight (grams) measurements.

- (a) 4 Marks Describe the relationship between weight and age shown in Figure 1.
- (b) 4 Marks Given four observations: Write the design matrix for a separate regression

Weight (g)	Age (weeks)	Location
0.44	3	1
0.50	3	1
1.76	3	2
2.38	4	2

model of Weight on Age and Location.

- (c) 2 Marks Write the two regression lines (one per location) for the separate model.
- (d) 6 Marks Estimate the weight of an 8-week-old mussel at Location 2.
- (e) 4 Marks Test for a significant interaction term ($\alpha = 0.05$), stating:
- Hypotheses
 - Test statistic
 - P-value
 - Conclusion
- (f) 6 Marks Using BIC, determine which model (separate or parallel) fits best.
- (g) 4 Marks Assess the linear model assumptions using Figure 2.

Appendix

Load the data

```
library(tidyverse)
mussel <- read.csv("mussel.csv", header=T)
mussel$Location <- factor(mussel$Location)
```

Visualise data

```
ggplot(mussel, aes(x = Age, y=Weight, col = Location,
shape = Location, fill=Location)) +
geom_point(size=3, alpha=0.5) +
geom_smooth(method = "lm", col = "black", aes(linetype=Location)) +
scale_linetype_manual(values=c("1"="solid", "2"="dashed")) +
labs(x = "Age (weeks)", y = "Weight (grams)", col="Location",
      shape="Location", fill="Location", linetype="Location") +
theme(legend.position = "top")
```

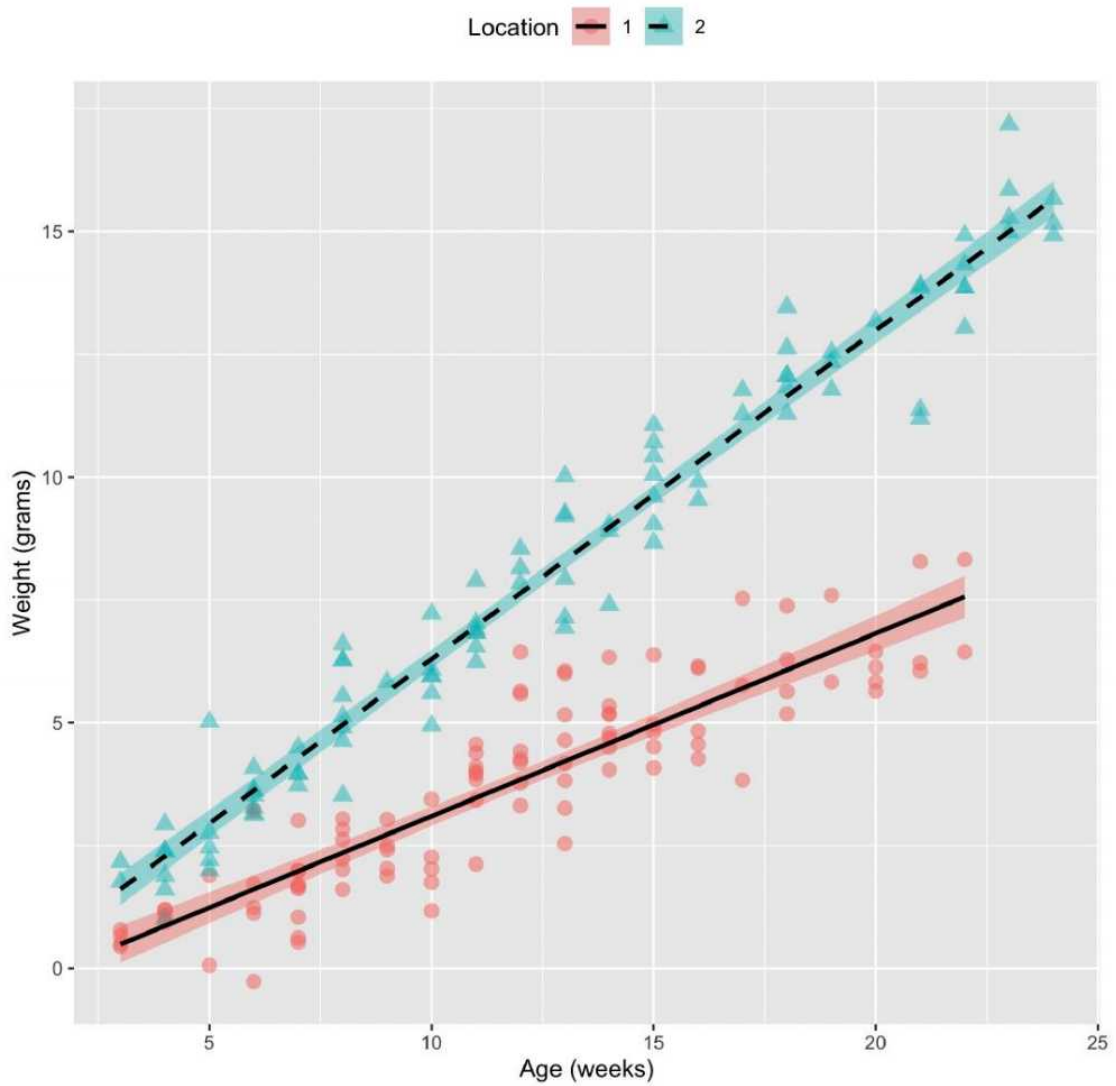


Figure 1: Scatterplot of Weight against Age for the mussel dataset. Colour and shape of points indicates Location.

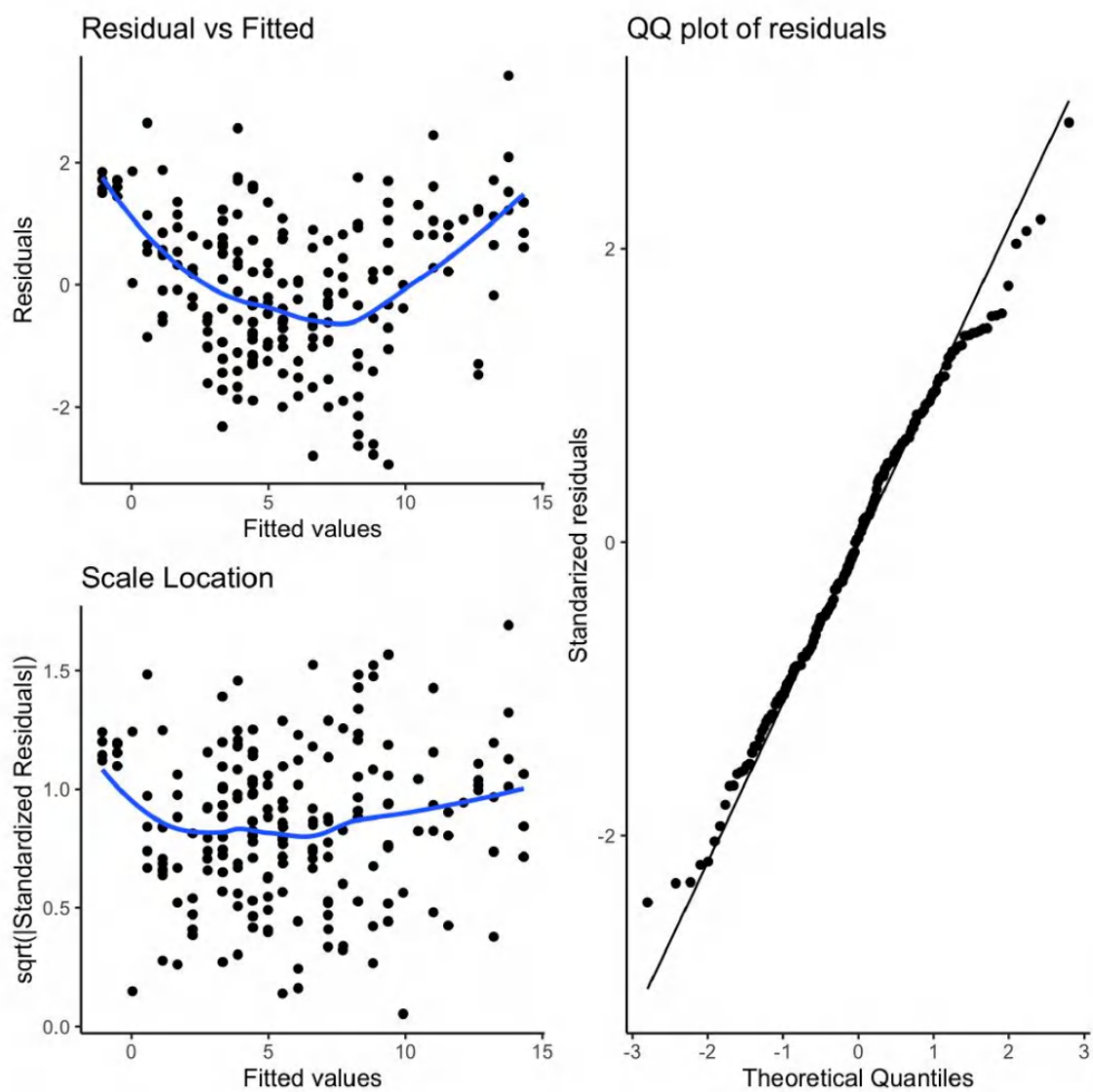


Figure 2: Assumption plots of the separate model for the mussel dataset.